PAPERGRAPH / COMPLETE TYPED ARGUMENT ATLAS

92 source-bound nodes · 86 declared links · 7 AND groups · no added observations

This is a zoomable audit figure, not an A4 summary. Role columns are layout only. Isolated nodes remain isolated when no semantic edge is declared.

requires · defines · joint support · motivates · reading order · supports within · does not establish · counterexample to interchangeability · qualifies · distinguishes scope from · derives under

Evidence facts 1

F_raw
Primary raw records
177,006 sequencing records

F_reference_changes
Resolved native-event labels changed by the reference stress test
137 record identifiers

F_jobs
Historical newly executed alignment jobs reported in the article
144 alignment jobs

F_streams
Nonduplicate streams in the full observer audit
15 streams

F_prefix
Interrupted literal anchor prefixes in the conditioned subset
Structured data: inspect model

F_geometry_target
Target PE-minus-mock native local event hull
Structured data: inspect model

F_unknown_partition
Disjoint reasons for the primary witness unknown status
Structured data: inspect model

F_witness_contrast_3
Native local witness contrast in CFB-06-04
Structured data: inspect model

F_withdrawal_1
Minimum hypothetical individual-label withdrawal in CFB-26-16
Structured data: inspect model

F_family_withdrawal
Whole-coding-family sensitivity in the native PE-minus-mock comparisons
Structured data: inspect model

F_grid_min_3
Minimum native projected PE-minus-mock lower endpoint in CFB-06-04
25.16093372795812 percentage points

F_stock
Stock-profile example changes the software statistic
Structured data: inspect model

F_joint_outer
Looser outer hull obtained by subtracting marginal hulls
Structured data: inspect model

F_external_NHEJ
External catalog: NHEJ
Structured data: inspect model

F_factorization_checks
Deterministic representation-target checks
90 overlapping checks

F_projection_replay
Per-reference labels checked by the portable projection checker
354,012 per-reference record labels

F_config_discrepancy
Preserved EMX1 documentation-versus-report configuration discrepancy
Structured data: inspect model

Evidence facts 2

F_witness
Resolved common-string witness records
170,725 sequencing records

F_unequal_strings
Unequal coding strings among jointly resolved records
83 record identifiers

F_external_rows
Actual external assignment-table rows
10,809 table rows

F_joint_resolved
Records jointly resolved by the compared coding extractions
170,808 sequencing records

F_geometry_edge
Edge PE-minus-mock native local event hull
Structured data: inspect model

F_rank_adverse
Small operator ranking changes with quality policy
Structured data: inspect model

F_witness_contrast_1
Native local witness contrast in CFB-26-16
Structured data: inspect model

F_worked_treated
Worked native local witness partition: PE treated
Structured data: inspect model

F_withdrawal_2
Minimum hypothetical individual-label withdrawal in CFB-26-02
Structured data: inspect model

F_grid_min_1
Minimum native projected PE-minus-mock lower endpoint in CFB-26-16
16.01852147591688 percentage points

F_grid_profiles
Finite parameter execution structure
Structured data: inspect model

F_joint
Resolved joint CFTR signatures
Structured data: inspect model

F_repeat_examples
Inspectable records with different extracted coding lengths
Structured data: inspect model

F_external_P23H_WT
External catalog: P23H/WT
Structured data: inspect model

F_emx1
Constant targets in the EMX1 interpretive control
Structured data: inspect model

F_external_replay
External computation reconstruction scope
Structured data: inspect model

Evidence facts 3

F_unknown
Unknown records retained inside the raw denominator
6,281 sequencing records

F_equal_labels_unequal_strings
Equal event labels despite unequal coding strings
18 record identifiers

F_all_records
Records in the full deduplicated observer audit
429,812 sequencing records

F_geometry
Documented edge-anchor geometry
Structured data: inspect model

F_geometry_inner
Inner PE-minus-mock native local event hull
Structured data: inspect model

F_truncation
Eight-base truncation and original ambiguity veto
Structured data: inspect model

F_witness_contrast_2
Native local witness contrast in CFB-26-02
Structured data: inspect model

F_worked_mock
Worked native local witness partition: mock
Structured data: inspect model

F_withdrawal_3
Minimum hypothetical individual-label withdrawal in CFB-06-04
Structured data: inspect model

F_grid_min_2
Minimum native projected PE-minus-mock lower endpoint in CFB-26-02
29.057937551051584 percentage points

F_grid_conflicts
Distinct resolved-label conflicts across profile families
Structured data: inspect model

F_joint_bounds
Native non-exact joint event hull
Structured data: inspect model

F_mask_witnesses
External source labels and complete-string differences use different scopes
Structured data: inspect model

F_external_EMX1
External catalog: EMX1
Structured data: inspect model

F_reporting
Reporting-contract exercise
Structured data: inspect model

F_external_raw_gaps
Raw input units outside the aligned catalogs
Structured data: inspect model

Declared premises

A_universe
Same authenticated raw universe

A_event
Same event and sequence contract

A_quality
Shared original quality policy

A_labels
Accepted labels are correct

A_completion
Unrestricted combinatorial unknown completion

A_partition
Valid integer partition

A_projection
Pinned projection adapter

A_witness
Three resolved identical strings

A_grid
Finite declared configuration family

A_catalog
Finite actual weighted catalog

A_target_resolved
Resolved declared target

A_timing
Declared information timing

A_source_identity
Source and implementation identity

Formal results

D_anchor_containment
Complete-anchor containment

D_finite_hull
Sharp finite completion hull

D_factorization
Observed resolved-catalog factorization criterion

D_refinement
Compatible information refinement narrows the hull

D_witness_containment
A conservative accepted subset widens the hull

D_withdrawal
Capacity-constrained minimum label withdrawal

R_equal_labels_imply_equal_strings
REFUTED INFERENCE: Equal event labels imply equal coding strings

Joint requirements

H_geometry
The local geometry interpretation combines actual coordinates and conditioned diagnostics under the shared comparison contract; it does not attribute every loss causally.

H_hull
The exact finite theorem requires the declared integer partition and completion premises.

H_witness
The actual conservative accepted domain is bound to all three resolved identical observed strings and its recorded count partition.

H_positive
Actual count partitions, exact positive lower endpoints and the conditional finite hull rule jointly support the stated positive witness contrasts.

H_grid
Actual finite-family extrema and record conflicts support the separate retrospective grid statement under its declared contract.

H_factorization
The classical iff criterion applies to the declared fixed resolved finite catalog; a certificate additionally requires checking actual fiber constancy.

H_reporting
The case-exercised reporting proposal requires the declared fields and actual exercise record; usability or community adoption is not derived.

Article stations

PG01
Start with the question

PG02
Declare the contract

PG03
Product → observability

PG04
Derive the bounds

PG05
Inspect correspondence

PG06
Construct the witness

PG07
Read the conclusion

PG08
Stress the calculation

PG09
Derive representation limits

PG10
Use diagnostic counterexamples

PG11
Audit the support

PG12
Consequences and next tests

Unestablished claims

L_accuracy
Comparative molecular accuracy

L_function
Product → RNA → protein → function and injury

L_transport
Prospective transport

L_usability
PaperGraph reader benefit

Grey dotted reading-order links carry no inference. Dashed premise links must be taken jointly at their named AND group.

Full source identities, exact quantities, conditions and objects are retained in the embedded JSON metadata and in the companion model.